ANALYSTLAB AFRICA

Experience Lab Internship Programme

WEEK 5: HEALTHCONNECT SOLUTION DEVELOPMENT & IMPLEMENTATION

Data Analytics Track — Initial HealthConnect Analytics Report

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Part 1: Week 4 Foundation Review

This section briefly restates the Week 4 foundation this Week 5 submission builds directly on.

1. Problem Defined

HealthConnect Clinic loses a significant share of booked appointment slots to missed attendance. The core question guiding this work is: how can HealthConnect use data to understand and reduce missed appointments (no-shows) while improving patient support?

2. Week 4 Proposed Approach

Perform structured exploratory analysis on the HealthConnect Appointment Dataset (5,000 records) to identify the strongest behavioural and operational predictors of no-shows — including reminder usage, prior no-show history, booking lead time, distance to clinic, and appointment scheduling factors — then translate these into KPIs and business recommendations.

3. Relevant Resources

- HealthConnect_Appointment_Data.csv — 5,000 anonymised appointment records
- HealthConnect_Data_Dictionary.xlsx — variable definitions and types

4. Key Assumptions & Limitations Identified in Week 4

- Dataset is fictional/synthetic — patterns are illustrative, not clinically validated
- No seasonal or holiday-calendar data available to explain date-based fluctuations
- Appointment outcome categories (Attended, No-Show, Cancelled) are assumed accurately recorded at source

5. Proposed Focus for Week 5

Move from planning into practical implementation: clean and validate the dataset, run exploratory analysis, calculate KPIs, build an analytical dashboard, and produce actionable business insights.

Part 2: Data Preparation Summary

The dataset (5,000 records, 18 relevant variables) was reviewed for data-quality issues before analysis. No duplicate appointment_id values were found.

Missing Values

distance_to_clinic_km: 90 missing values (1.8% of records). No logical relationship exists between distance and appointment outcome, so all 90 were imputed using the column median (8.7 km).

waiting_time_minutes: 60 missing values (1.2% of records). Missing values were cross-checked against appointment_outcome to distinguish structural gaps from genuine data-quality issues. Results: 37 belonged to No-Show appointments, 2 to Cancelled appointments, and 21 to Attended appointments. The 39 No-Show/Cancelled values were left blank, since these patients never attended the clinic and could not have accrued a waiting time — expected, structural missingness rather than an error. The remaining 21 Attended values represented genuine missing data and were imputed using the column median (24 minutes).

This mixed approach avoided distorting the dataset with implausible values (e.g. assigning a wait time to a patient who never attended) while ensuring completeness where the missing data was analytically meaningful.

Other Observations

- reminder_channel shows 1,366 blanks — these correspond exactly to records where reminder_sent = "No", so this reflects "not applicable" rather than missing data.

- All date, numeric, and categorical fields were confirmed to match the expected data types in the Data Dictionary.
- No duplicate appointment_id values were found across the 5,000 records.

Part 3: Exploratory Data Analysis

Appointment attendance and no-show patterns were investigated across reminder usage, prior appointment history, booking lead time, distance to clinic, gender, appointment day, and appointment time. Key relationships are summarised below and visualised in the accompanying Power BI dashboard.

- Overall attendance split: 2,314 Attended, 2,423 No-Show, 263 Cancelled (out of 5,000).
- Patients who did not receive a reminder no-showed at a higher rate (51.39%) than those who did.
- Patients with at least one prior no-show missed appointments at 55.41%, versus a markedly lower rate for first-time patients.
- Average booking lead time for no-shows (34.53 days) was notably higher than for attended appointments, suggesting appointments booked far in advance are more likely to be forgotten or deprioritised.
- No-show rate varies modestly by gender (Male 49%, Female 48%, Prefer not to say 44%) and by appointment time of day (Evening 49.78%, Afternoon 48.38%, Morning 48.14%).
- No-show rate varies by reminder channel, with clear differences between SMS, Email, WhatsApp, and no reminder at all.

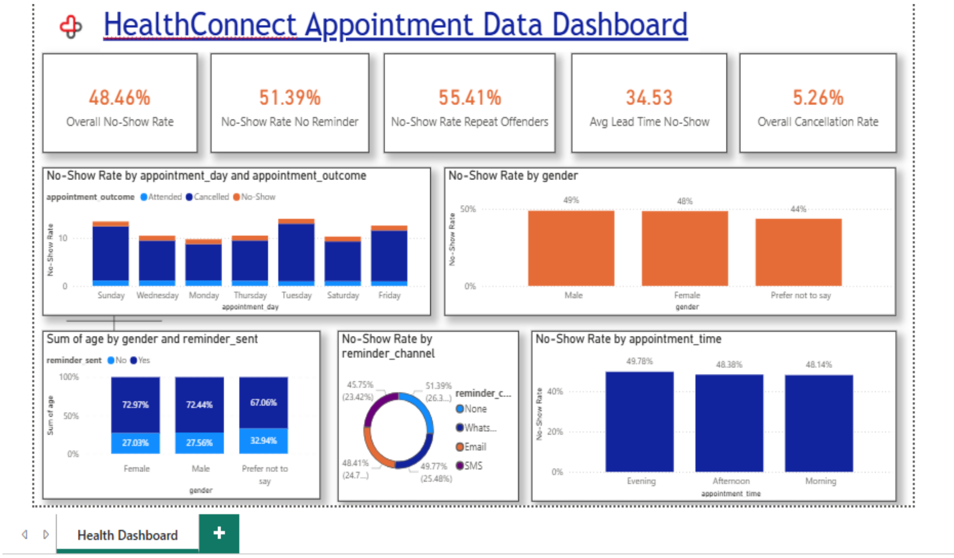
Part 4: KPI Development

Five KPIs were selected from the Week 4 candidate list, calculated in both Excel and Power BI (DAX measures), and validated for consistency across both tools.

KPI	Value & Interpretation
Overall No-Show Rate	48.46% of all appointments end in a no-show. This is the baseline measure of the scale of the problem HealthConnect needs to solve.
No-Show Rate — No Reminder	51.39% no-show rate when no reminder is sent, versus a lower rate when a reminder is sent. Confirms reminders reduce no-shows, though the effect is modest — reminders alone will not solve the problem.
No-Show Rate — Repeat Offenders	55.41% no-show rate among patients with a prior no-show history, notably higher than first-time patients. This is the strongest predictor identified and the clearest target for intervention.
Average Lead Time — No-Show	34.53 days average booking lead time for no-shows, versus a shorter lead time for attended appointments. Longer lead times correlate with higher no-show risk.
Overall Cancellation Rate	5.26% of appointments are properly cancelled rather than missed silently — a much healthier outcome than a no-show, since the slot can be reallocated.

Part 5: Analytical Dashboard

An interactive Power BI dashboard was built containing the 5 KPI cards above alongside supporting visuals: No-Show Rate by appointment day, No-Show Rate by gender, No-Show Rate by reminder channel, No-Show Rate by appointment time, and age/gender/reminder breakdowns. The dashboard file (.pbix) is included as a supporting file with this submission.



Part 6: Business Insights

- Nearly half of all appointments are missed (48.46% no-show rate) — HealthConnect is losing close to 1 in 2 booked slots, with direct revenue and care-continuity implications.
- Reminders help, but only modestly — no-show rate drops when a reminder is sent versus not sent, meaning reminders should continue but cannot be the only intervention.
- Past behaviour predicts future no-shows — patients with previous no-shows miss appointments at 55.41%, well above first-timers. This group should be prioritised for extra outreach (calls, double reminders, deposit-style commitments).
- Patients who book far in advance are more likely to no-show (avg. 34.53 days lead time vs. shorter for attended) — a follow-up reminder closer to the appointment date could reduce forgetfulness.
- Gender shows a mild difference in no-show behaviour (Male 49% vs Female 48% vs Prefer not to say 44%) — not a priority lever, but worth monitoring.
- Reminder channel matters — no-show rates differ by channel (SMS, Email, WhatsApp, None); HealthConnect should standardise on whichever channel shows the lowest no-show rate.

Part 7: Cross-Track Collaboration

Track interacted with: Data Science.

Information exchanged: Key EDA findings were shared — specifically that previous_no_shows, reminder_sent, and booking_lead_days show meaningful relationships with appointment_outcome.

Why relevant: These variables are strong candidates for inclusion as predictive features in a no-show classification model.

What changed as a result: The Data Science track's Week 5 feature engineering can prioritise these fields rather than testing all variables from scratch, saving development time and grounding their model in validated relationships.

Part 8: Assumptions, Limitations, Risks & Dependencies

Still Relevant

- Dataset remains fictional/synthetic — findings are illustrative, not clinically validated.
- No seasonal or holiday-calendar data is available to explain date-based fluctuations.

Resolved in Week 5

- Missing values in distance_to_clinic_km and waiting_time_minutes were identified, investigated, and resolved (median imputation / structural-gap treatment).
- Data types and duplicate records were verified as clean.

New Issues Discovered

- reminder_channel is blank whenever no reminder was sent — this is expected, not a data-quality defect, but should be documented for downstream tracks (e.g. Generative AI, Data Science) to avoid misinterpretation.

Impact & Next Steps

None of the above materially affect the reliability of the Week 5 KPIs or insights. The main risk carried into Week 6 is the synthetic nature of the dataset, which should be caveated in any stakeholder-facing summary.

Part 9: Week 5 Project Summary

1. Planned for Week 5

Move from Week 4 planning into practical implementation: clean the dataset, run EDA, calculate KPIs, build a dashboard, and produce business insights.

2. Actually Completed

Full data-quality review and cleaning, exploratory data analysis across six key variables, five KPIs calculated and cross-validated in Excel and Power BI, a six-visual analytical dashboard, six business insights, and cross-track documentation.

3. Key Findings

A 48.46% baseline no-show rate driven most strongly by prior no-show history (55.41%) and long booking lead times (34.53 days average for no-shows), with reminders providing a modest but real protective effect.

4. Major Challenges

Determining the correct treatment for missing waiting_time_minutes values required cross-checking against appointment outcome to distinguish genuine missing data from expected structural gaps, rather than applying a blanket imputation.

5. Key Decisions & Why

Used a mixed imputation strategy for waiting_time_minutes (median for Attended rows only, left blank for No-Show/Cancelled) rather than a single blanket median fill, to avoid implying wait times for patients who never attended.

6. Changes to Week 4 Approach

No changes to the core approach; Week 5 executed the analysis exactly as scoped in Week 4.

7. Cross-Track Collaboration Completed

Shared key predictive-relationship findings with the Data Science track to inform their feature engineering (see Part 7).

8. Remaining Work

Deeper segmentation (e.g. combined effect of reminder channel and lead time), and refinement of the dashboard's remaining visuals for Week 6.

9. Proposed Focus for Week 6

Build on these findings to support the Data Science track's predictive modelling work and refine recommendations into a stakeholder-ready summary.